SOMAIYA VIDYAVIHAR UNIVERSITY -- K J SOMAIYA COLLEGE OF ENGINEERING

Department of Artificial Intelligence & Data Science | Academic Year 2026-2027

STATISTICAL INVESTIGATION SHEET (IA-1)

Course: Statistical Methods & Probability (SMP) **Theme:** STAT-AI Engineering Challenge	**Case Study 1:** AI Heatwave Monitoring System **Evaluation:** 20 Marks (Live Showcase & Viva)	**Format:** GroupNumber_CaseStudy1_StatAI_IA1 **Live App:** github.com/varad-op/smp-maths-app

1. Problem Statement & Engineering Motivation

Heatwaves represent one of the most hazardous meteorological extremes in India, leading to acute heat stroke admissions, agricultural stress, municipal water shortages, and electrical power grid surges under peak cooling loads. Conventional monitoring often relies on retrospective or isolated manual weather alerts without probabilistic grounding. The objective of our engineering investigation is to design an **AI-Supported Statistical Early Warning System** that continuously ingests daily surface temperature and humidity telemetry, performs rigorous descriptive and inferential statistical modeling, and automatically transitions between municipal action levels (**Green, Yellow, Orange, Red**) to enable pre-emptive disaster mitigation across hospital triage, labor directives, water supply, and power grids.

2. Description of the Meteorological Dataset

• **Source of Data:** India Meteorological Department (IMD) synoptic station archives (Santacruz/Colaba) cross-referenced with NASA POWER Surface Meteorology API for the Mumbai & inland Maharashtra transition corridor.
• **Observation Count (N):** 65 sequential daily observations covering the pre-monsoon summer spell (March 15 to May 18, 2026).
• **Recorded Parameters:** Date (YYYY-MM-DD), Max_Temp_C (°C), Min_Temp_C (°C), Relative_Humidity_Pct (%), Heat_Index_C (°C, NOAA Rothfusz apparent temperature), Weather_Condition (Categorical IMD status).
• **Data Hygiene & IQR Outlier Screening:**
Q1 = 35.90 °C, Median = **37.30 °C**, **Q3 = 39.00 °C** | **IQR = Q3 - Q1 = 39.00 - 35.90 = 3.10 °C**
Lower Fence = Q1 - 1.5(IQR) = **31.25 °C** | **Upper Fence** = Q3 + 1.5(IQR) = **43.65 °C**
Outlier Assessment: Two observations exceed the upper fence: **44.40 °C** (May 6) and **43.80 °C** (May 7). Cross-verification with IMD synoptic charts confirms these are genuine meteorological extremes caused by an advective continental heat plume, rather than sensor transmission errors. They are intentionally retained to preserve true extreme tail behavior.

3. Statistical Concepts Used & Relevant Formulae

A. Data Hygiene & Dispersion:
IQR = Q3 - Q1 | Inner Fences = [Q1 - 1.5(IQR), Q3 + 1.5(IQR)]
Sample Mean: x_bar = (Sum x_i) / N | Sample Variance: s^2 = Sum(x_i - x_bar)^2 / (N - 1)
Sample Std Dev: s = sqrt(s^2) | Coefficient of Variation: CV = (s / x_bar) * 100%
Grouped Mean: x_grouped = (Sum f_i * x_i) / (Sum f_i) | Grouped Median: M_d = L + [(N/2 - cf_prev)/f] * h
B. Gaussian Normal Distribution Modeling:
Assuming X ~ N(mu, sigma^2): Standard Normal Variable: Z = (X - mu) / sigma
Tail Exceedance Probability: P(X >= T) = 1 - Phi(Z) = Integral [1 / (sigma * sqrt(2*pi))] * exp[-0.5((x-mu)/sigma)^2] dx
C. Joint Contingency & Bayesian Inference:
Conditional Probability: P(Heatwave | High Humidity) = n(Heatwave and High Humidity) / n(High Humidity)
Bayes' Theorem for Sensor Alarm: P(H | S) = [P(S | H) * P(H)] / [P(S | H)*P(H) + P(S | not-H)*P(not-H)]
D. Discrete Random Process & Autocorrelation:
Sample ACF at Lag k: r_k = [Sum (X_t - x_bar)(X_{t+k} - x_bar)] / [Sum (X_t - x_bar)^2], for k in {1, 2, ..., 7}
Bartlett's 95% Confidence Bounds for White Noise: CI_95% = ± 1.96 / sqrt(N)

4. Step-by-Step Numerical Calculations (Derived from Dataset)

Calculation 1: Descriptive Statistics & Central Tendency (N = 65)

From our 65 daily observations of Maximum Temperature (T_max):
• Total Sum: Sum x_i = **2,439.40 °C**
• Sample Mean: x_bar = 2,439.40 / 65 = **37.53 °C**
• Sample Median: M_d = 33rd observation in sorted series = **37.30 °C** | Mode: M_o = **36.80 °C**
• Sum of Squared Deviations: Sum(x_i - x_bar)^2 = **476.12**
• Sample Variance: s^2 = 476.12 / (65 - 1) = 476.12 / 64 = **7.4393 (°C)^2**
• Sample Standard Deviation: s = sqrt(7.4393) = **2.73 °C**
• Coefficient of Variation: CV = (2.73 / 37.53) * 100% = **7.27%**
Engineering Interpretation: The relatively low CV (7.27%) indicates that maritime sea breezes stabilize baseline temperatures around 37-38°C; therefore, excursions past 40°C represent statistically significant high-sigma deviations.

Calculation 2: Continuous Frequency Distribution (7 Structured Bins)

Class Interval (°C)	Class Mark (x_i)	Frequency (f_i)	Cum Freq (cf)	Rel Freq (%)	f_i * x_i	f_i * (x_i)^2
31.0 - 33.0	32.0	3	3	4.62%	96.00	3,072.00
33.0 - 35.0	34.0	6	9	9.23%	204.00	6,936.00
35.0 - 37.0	36.0	17	26	26.15%	612.00	22,032.00
37.0 - 39.0	38.0	23	49	35.38%	874.00	33,212.00
39.0 - 41.0	40.0	10	59	15.38%	400.00	16,000.00
41.0 - 43.0	42.0	2	61	3.08%	84.00	3,528.00
43.0 - 45.0	44.0	4	65	6.15%	176.00	7,744.00
Total / Sum	**--**	**N = 65**	**--**	**100.00%**	**Sum = 2,446.00**	**Sum = 92,524.00**

• **Grouped Mean Calculation:**
x_grouped = (Sum f_i * x_i) / N = 2,446.00 / 65 = **37.63 °C**
• **Grouped Median Calculation:**
Median class contains N/2 = 32.5 observation in [37.0 - 39.0 °C] with L = 37.0, cf_prev = 26, f = 23, h = 2.0:
M_d = 37.0 + [(32.5 - 26) / 23] * 2.0 = 37.0 + [6.5 / 23] * 2.0 = 37.0 + 0.565 = **37.57 °C**
• **Discretization Discrepancy:** Ungrouped Mean (37.53 °C) vs Grouped Mean (37.63 °C) exhibits a difference of only Delta = **+0.10 °C (0.27%)**, proving that 7 uniform bins preserve data distribution with negligible error.

Calculation 3: Extreme Value Probability via Normal Distribution

Fitting a Gaussian model to daily maximum temperature: **X ~ N(mu = 37.53 °C, sigma = 2.73 °C)**.
Standardized Z-score Z = (T - 37.53) / 2.73 and tail risk P(X >= T) = 1 - Phi(Z):
• **Threshold T = 38.0 °C (Advisory Watch Level):**
Z_38 = (38.0 - 37.53) / 2.73 = **+0.1722** | P(X >= 38.0 °C) = 1 - Phi(0.17) = 1 - 0.5683 = **43.17%**
• **Threshold T = 40.0 °C (Official IMD Heatwave Criteria):**
Z_40 = (40.0 - 37.53) / 2.73 = **+0.9048** | P(X >= 40.0 °C) = 1 - Phi(0.90) = 1 - 0.8172 = **18.28%**
• **Threshold T = 42.0 °C (Severe Heatwave Emergency):**
Z_42 = (42.0 - 37.53) / 2.73 = **+1.6374** | P(X >= 42.0 °C) = 1 - Phi(1.64) = 1 - 0.9492 = **5.08%**
• **Threshold T = 45.0 °C (Catastrophic Extreme):**
Z_45 = (45.0 - 37.53) / 2.73 = **+2.7363** | P(X >= 45.0 °C) = 1 - Phi(2.74) = 1 - 0.9969 = **0.31%**
Engineering Value: An 18.28% likelihood of exceeding 40°C means heatwaves occur roughly 1 out of every 5.5 summer days, justifying permanent standby municipal heat protocols rather than reactive ad-hoc management.

Calculation 4: Joint Contingency Matrix & Bayesian Alarm Filtering

Temperature Category	High Humidity (>60%)	Normal Humidity (≤60%)	Marginal Total
Heatwave (≥ 40.0 °C)	0 days (0.0%)	7 days (10.8%)	**7 days (10.77%)**
Moderate (< 40.0 °C)	40 days (61.5%)	18 days (27.7%)	**58 days (89.23%)**
Marginal Total	**40 days (61.5%)**	**25 days (38.5%)**	**N = 65 (100.0%)**

• Prior Heatwave Probability: P(H) = 7 / 65 = **10.77%** | Prior Non-Heatwave: P(not-H) = 58 / 65 = **89.23%**
• *Coastal Meteorological Anomaly:* P(Heatwave | High Humidity) = 0 / 40 = **0.00%**. Severe dry heat extremes in Mumbai occur exclusively under continental north-easterly wind advection; sea breezes import humidity but suppress ambient temperature below 38°C.
• **Bayesian Update for Automated IoT Sensors:** Given Sensor Sensitivity P(S | H) = 0.92 and False Alarm Rate P(S | not-H) = 0.08:
Marginal Alarm Probability: P(S) = (0.92 * 0.1077) + (0.08 * 0.8923) = 0.0991 + 0.0714 = **0.1705**
Posterior Heatwave Probability: P(H | S) = [0.92 * 0.1077] / 0.1705 = 0.0991 / 0.1705 = **0.5812 (58.12%)**
Engineering Value: When the automated sensor fires, our confidence jumps from 10.77% to 58.12%, successfully eliminating 41.88% of false alerts before mobilizing expensive municipal water tankers and medical teams.

Calculation 5: Autocorrelation (ACF) & Streak Duration

Modeling daily sequence as discrete time series {X_t}. For N = 65, **Bartlett's 95% Confidence Bound** = ± 1.96 / sqrt(65) = **± 0.2431**.
• **Autocorrelation Coefficients:** Lag 1 (r1) = **+0.9026** (Significant) | Lag 2 (r2) = **+0.8050** (Significant)
Lag 3 (r3) = **+0.6862** (Significant) | Lag 4 (r4) = **+0.5291** (Significant) | Lag 5 (r5) = **+0.3717** (Significant)
Lag 6 (r6) = **+0.1926** (Insignificant) | Lag 7 (r7) = **+0.0455** (Insignificant, memory dissipates)
• **Heatwave Streak:** We detect a continuous **6-day heatwave spell (May 4-9, 2026)** with peak temperature reaching **44.40 °C**.
Inference: The strong positive autocorrelation up to Lag 5 proves daily heat is not memoryless white noise. Multi-day thermal accumulation creates compounded human physiological stress, requiring escalated cumulative alert responses.

5. AI Decision Engine & Mandated Municipal Directives

Alert Tier	Thermal Condition	Statistical Indicator	Mandated Engineering & Civil Directives
GREEN ALERT (Normal)	Effective Temp < 38.0 °C Heat Index < 45.0 °C	P(T ≥ 40) < 15% Normal baseline	• Standard automated meteorological telemetry logging active. • Baseline water reservoir and electrical grid operation.
YELLOW ALERT (Advisory)	38.0 °C ≤ Temp < 40.0 °C or Hazard Score ≥ 40	P(T ≥ 40) appx. 18% Early thermal stress	• Issue public health advisories on municipal portals and transit radios. • Primary clinics on standby (+30 beds triage). Commercial AC to 24°C.
ORANGE ALERT (Heatwave)	40.0 °C ≤ Temp < 42.5 °C or HI ≥ 50 °C (or Streak ≥ 2d)	P(T ≥ 40) > 30% or P(H|S) > 55%	• Restrict primary school hours until 11:00 AM; cancel outdoor sports. • Deploy transit water kiosks ('Pyaaos'). Hospitals add +85 beds with ORS.
RED ALERT (Severe Emergency)	Temp ≥ 42.5 °C or HI ≥ 55 °C or Hazard Score ≥ 80	P(T ≥ 42) > 5% Acute life hazard	• Mandatory labor ban on outdoor construction from 11:30 AM to 4:00 PM. • Activate dedicated heatstroke ICU (+180 beds). Ramp grid reserves (+32%). • Mobilize emergency water tankers (+40% volume) to informal settlements.

6. Statistical Limitations of the Model

1. **Gaussian Tail Asymmetry:** While surface air temperatures approximate a bell curve near the mean, extreme heat events exhibit positive right-tail skewness. Generalized Extreme Value (GEV/Gumbel) distributions provide higher fidelity for decadal tails.
2. **Seasonal Non-Stationarity:** Meteorological processes transition from pre-monsoon dry heat to monsoonal humidity; static parameter estimates underestimate seasonal drift.
3. **Microclimatic Urban Heat Island (UHI) Gradients:** Single regional telemetry fails to reflect asphalt canyons that experience +1.2 °C to +2.5 °C over green buffers.

7. Authoritative References

1. **India Meteorological Department (IMD)**, Government of India: *Criteria for Declaring Heat Wave and Severe Heat Wave Conditions*, NWFC (2024).
2. **National Disaster Management Authority (NDMA)**, Government of India: *National Guidelines for Preparation of Action Plan - Prevention of Heat Wave* (2022).
3. **National Oceanic and Atmospheric Administration (NOAA) / NWS**: *The Heat Index Equation and Rothfusz Regression*, Technical Attachment (2021).
4. **Montgomery, D. C., & Runger, G. C.**: *Applied Statistics and Probability for Engineers*, 7th Edition, John Wiley & Sons.